
Re-Implementing CNN Price-Trend Signals: A Reproduction of Jiang, Kelly, and Xiu (2023)

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Abstract

This paper reproduces the image-based price-trend prediction framework of Jiang et al. [2023] (hereafter the *original paper*), which replaces predefined technical indicators with OHLC-volume charts and convolutional neural networks (CNNs). Using CRSP daily U.S. equity data, we reconstruct the image pipeline, train horizon-specific CNN ensembles, and evaluate the resulting signals through out-of-sample backtests. We compare the CNN signals with momentum, moving-average, and reversal benchmarks, and assess residual performance through factor-spanning regressions. The reproduction is strongest at short horizons, where image-based signals deliver the best economic performance. Relative to the magnitudes reported by the original paper, medium- and long-horizon strategies weaken substantially after costs, and factor stripping yields significantly negative alphas. Saliency maps indicate that the CNN mainly attends to recent price movement and volume contrast. We also provide replication materials so that the empirical results can be audited and extended.

1 Introduction

Technical analysis has long suggested that the visual “shape” of a price chart conveys information beyond what is captured by scalar momentum or moving-average filters. The original paper formalises this intuition by feeding fixed-resolution open–high–low–close (OHLC) images with a volume sub-panel into a convolutional neural network and showing that the resulting forecasts dominate canonical price-trend signals out of sample. The study joins a growing empirical asset-pricing literature that uses flexible machine-learning predictors to map raw historical features into expected returns [Gu et al., 2020, Kelly et al., 2019], and it ties directly to early efforts at formalising chart patterns via kernel smoothing [Lo et al., 2000].

This paper re-implements the core pipeline from end to end and evaluates it under the same in-sample and out-of-sample calendar design as the original paper: training and validation draw from 1993–2000 with a 70/30 random sample split, and headline out-of-sample results are reported for 2001–2019. Our contributions are:

1. An independent OHLC-plus-volume image renderer and horizon-matched CNN stacks built from the dimensions, drawing rules, and layer geometry documented in the original paper—including the simple moving-average overlay, volume sub-panel, and vertical orientation convention—implemented from the original paper’s written specification and reported tensor dimensions.
2. A CRSP-style daily panel for data construction, with headline out-of-sample CNN evaluation as above; proportional transaction costs; matched-horizon price momentum, simple moving-average cross, and short-term reversal baselines; and a five-factor-plus-momentum spanning regression [Fama and French, 1993, 2015, Carhart, 1997].

3. Input-gradient saliency analysis [Simonyan et al., 2014] of the $I = 60$ ensemble, which highlights the right tail of each image and the volume bars as the dominant feature regions.
4. Code, trained weights, and post-aggregation CSV tables are released on GitHub, allowing every numerical claim to be regenerated.

The remainder of the paper is organised as follows. Section 2 positions our reproduction relative to the chart-pattern, deep-learning, and interpretability literatures. Section 3 details data sources and image construction. Section 4 describes the CNN architecture and training recipe. Section 5 reports empirical results (decile prediction accuracy, baseline comparisons, cumulative returns, classification metrics, decile portfolio statistics, and Fama–French spanning regressions). Section 6 visualises model attention. Section 7 discusses where our results align with and diverge from the original paper.

2 Related work

Chart-pattern literature. Lo et al. [2000] provided one of the first systematic treatments of technical chart patterns by combining kernel regression with automated pattern recognition. Jegadeesh and Titman [1993] documented price momentum as a robust cross-sectional return predictor, providing the canonical baseline against which chart-based technical trend signals are often benchmarked empirically.

Deep learning for return prediction. Recent work demonstrates that deep architectures uncover return signals that are difficult to capture with closed-form characteristics. Gu et al. [2020] benchmark a broad set of supervised learners on U.S. equities; Kelly et al. [2019] use latent-factor models with characteristic loadings. Within technical analysis, Sezer and Ozbayoglu [2018] convert indicators into images for trading. The original paper unifies these threads by treating the OHLC chart itself as the model input.

Interpretability via input gradients. Simonyan et al. [2014] introduced gradient-based saliency to visualise which input pixels most influence a CNN’s class score; we apply the same technique to identify which segments of each OHLC image drive the model’s “up” probability. CNN architectures used here trace back to LeCun et al. [1998].

3 Data and image construction

Universe. We employ a CRSP-consistent daily security-level panel of U.S. equities spanning 1992-01-01, to 2024-12-31. Data are observed at the daily frequency for each security, uniquely identified by its CRSP permanent security number (PERMNO). The panel includes core trading and pricing fields: daily market capitalization, total and price returns, trading volume, and open-high-low-close (OHLC) prices. We impose a standard sequential screening procedure: all fields are required to be non-missing; we retain only securities with a closing price of at least \$1 and strictly positive trading volume. Log returns are computed, and OHLC prices are adjusted for corporate actions via cumulative log-return compounding, with the first valid observation for each security serving as the normalization anchor. No additional survivorship bias adjustments are applied beyond those inherent to the standard CRSP universe construction.

Labels and holding period. For each horizon $F \in \{5, 20, 60\}$ trading days we form $R_{t,F}^{\text{fut}} = \exp\left(\sum_{s=t+1}^{t+F} \log(1 + r_s)\right) - 1$ on the adjusted price path, and a binary classification label $y_{t,F} = \mathbf{1}\{R_{t,F}^{\text{fut}} > 0\}$. Rebalance timestamps are fixed at period-end calendar boundaries (Friday for $F = 5$, month-end for $F = 20$, quarter-end for $F = 60$), matching the canonical SPY trade calendar.

Image construction. Following the original paper, we encode each stock–date observation as a fixed-size grayscale *chart image* rather than a vector of technical indicators. The background is black (pixel value 0) and all price–volume objects are drawn in white (255). We consider three input horizons $I \in \{5, 20, 60\}$, each matched to the prediction horizon $F = I$, so the image summarizes the most recent I trading days of adjusted OHLC prices and daily volume.

OHLC panel. The upper portion of the image is reserved for open–high–low–close prices. Each trading day τ in the I -day window occupies three adjacent pixel columns: a one-pixel open marker in the left column, a vertical high–low bar in the centre, and a one-pixel close marker in the right column. Prices are linearly mapped into the vertical axis using the window minimum and maximum, where the range is defined to include both the raw OHLC values and the moving-average line described below. After drawing, the OHLC sub-panel is flipped top-to-bottom so that upward price movements read visually as “up” on the chart, in line with conventional price plots.

Moving-average line. We overlay a simple moving average (SMA) of the close price, using an I -day window—the same length as the chart horizon (e.g., a 20-day chart uses a 20-day SMA). For each day, the SMA value is placed on the centre column of that day; consecutive days are connected by a one-pixel poly-line. To ensure that the first in-window SMA already reflects I days of history, the SMA is computed from $2I - 1$ days of past closes ending at the chart date.

Volume sub-panel. The lower portion of the image displays volume bars. As in the original paper, the volume chart is drawn along the bottom of the image and occupies one-fifth of the total height; the OHLC panel fills the upper four-fifths, with a one-pixel gap between the two regions. Within the volume band, the largest daily volume in the window is scaled to the top of the band and all other days are drawn proportionally. Missing-volume days are left blank.

Image dimensions. Because each day uses three pixel columns, the image width is $W = 3I$. The total height H stacks the OHLC block, the separator, and the volume band. Table 1 reports the (W, H) pairs used in our implementation, identical to those in the original paper.

Table 1: Image Dimensions

Horizon	W	H
$I = 5$	15	32
$I = 20$	60	64
$I = 60$	180	96

Figure 1 provides a concrete illustration of the construction rules above. We plot the same security on the same calendar anchor for $I \in \{5, 20, 60\}$, so the only deliberate difference across panels is the look-back length. The left panel ($I = 5$) is the coarsest view: fifteen columns of price action and a compact volume strip. The centre and right panels progressively widen the canvas to sixty and one hundred eighty columns, revealing more fine-grained OHLC structure, a longer SMA trace, and richer intra-window variation in volume. Because every image fed to the CNN has fixed (W, H) for a given horizon, longer windows do not add pixels per trading day; instead, the model sees the same economic history at three standardised resolutions, analogous to viewing a price chart at weekly, monthly, and quarterly zoom levels.

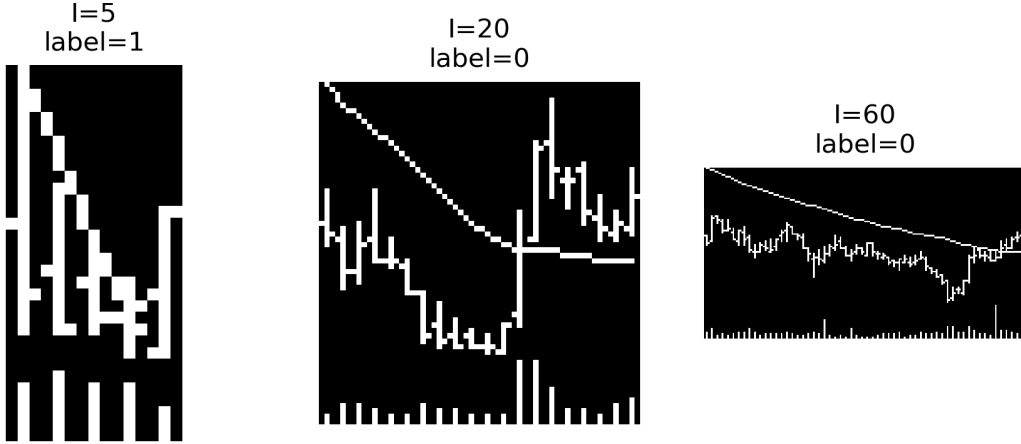


Figure 1: OHLC-plus-volume images for a single security at horizons $I \in \{5, 20, 60\}$.

Train/validation/test split. We follow the original paper in treating the early portion of the panel as in-sample. Training is performed on 1993–2000 with a 70/30 random sample-level split for training/validation. Out-of-sample backtests start in January 2001. Headline tables and figures report results through December 2019, matching the evaluation window in the original paper.

4 CNN architecture and training

Architectures. The three horizon-specific backbones mirror the published configuration. Each block applies a 5×3 convolution, LeakyReLU(0.01), batch normalisation, and 2×1 max pooling along the time axis. For $I = 20$ and $I = 60$, the opening convolution uses dilation and stride to widen the receptive field without a proportional increase in parameters, and pooling is configured so that spatial dimensions are preserved enough to yield the fully connected widths reported in the original paper; the $I = 5$ network is shallower and uses the default pooling rule throughout. Figure 2 summarises the three stacks; stride, dilation, and channel widths are listed in Table 5 (Appendix B).

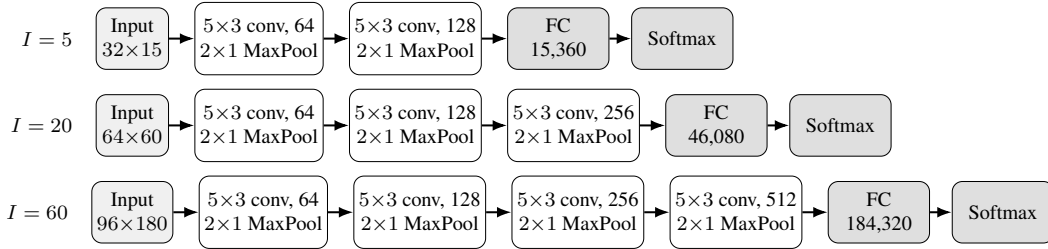


Figure 2: Horizon-specific CNN backbones. Each rectangle is one block: 5×3 convolution (with the channel width shown), LeakyReLU, batch normalisation, and 2×1 max pooling. Stride and dilation for $I = 20$ and $I = 60$ are in Table 5. Fully connected input sizes are 15,360 ($I = 5$), 46,080 ($I = 20$), and 184,320 ($I = 60$).

Optimisation. We minimise binary cross-entropy on the per-horizon label $y_{t,F}$ using stochastic gradient descent (Adam variant disabled), with learning rate 10^{-5} , batch size 128, and a maximum of 50 epochs with early-stopping on validation loss. Convolutional and linear weights are initialised with Xavier-uniform and biases at zero.

The original paper reports headline results from an ensemble of independently trained networks whose softmax outputs are averaged at inference. Our replication proceeded in two stages. For the course presentation, limited local compute allowed only a single random seed (42) per horizon, so we could not yet reproduce the full ensemble-averaging protocol. For the final project we reran training on rented cloud GPUs with five seeds $\{42, 123, 456, 789, 101112\}$ and aggregate by averaging softmax-probability outputs at inference time, as in the original paper. All empirical results in Sections 5–6 use this five-seed ensemble. The resulting model sizes are approximately 0.6 MB ($I = 5$), 2.8 MB ($I = 20$), and 11.8 MB ($I = 60$) per seed, consistent with the parameter counts implied by Figure 2.

5 Empirical results

We score every test-set observation with the five-seed ensemble and form ten cross-sectional deciles by predicted “up”-probability at each rebalance date. The high-minus-low (H–L) portfolio is equal-weighted within deciles, rebalanced at horizon-end, and adjusted for proportional transaction costs at 10 bp per side (20 bp round-trip) applied to both legs.

5.1 Decile prediction accuracy (Figure 6 reproduction)

At $I = 5$ the CNN decile annualised returns rise nearly monotonically from 6.62% in D1 to 20.67% in D10 (Figure 3), while the five-day momentum and moving-average curves are essentially flat across the cross-section. At longer horizons the picture changes: the $I = 20$ CNN spread between D10 and D1 is mildly positive (11.10% vs. 9.78%), and at $I = 60$ the CNN ordering is essentially flat (11.95% in D10 against 13.00% in D1); corresponding plots are provided in Appendix D. The

traditional baselines, by contrast, display a U-shaped pattern at $I = 20$ and $I = 60$: both the bottom and top deciles deliver above-average returns, reflecting the well-known concentration of momentum and mean-reversion premia in the extreme deciles.

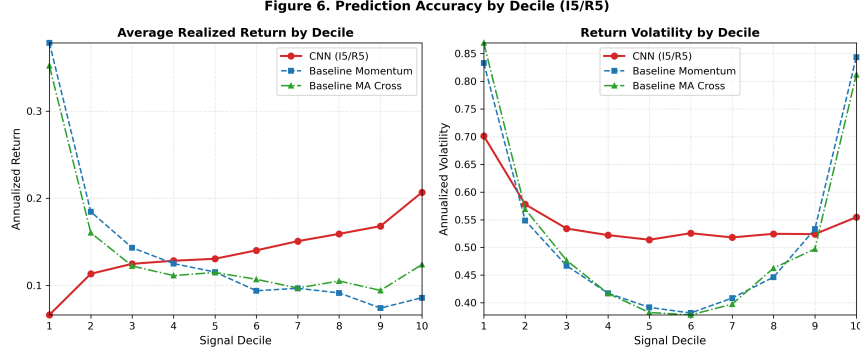


Figure 3: Annualised decile returns at the $I = 5$ horizon for the CNN, matched-horizon momentum, and moving-average cross signals. Reproduction of Figure 6(a) of the original paper; series are computed from the same out-of-sample prediction panels and rebalance calendar. Analogous plots for $I = 20$ and $I = 60$ are in Appendix D.

5.2 Comparison with traditional baselines (Figure 7)

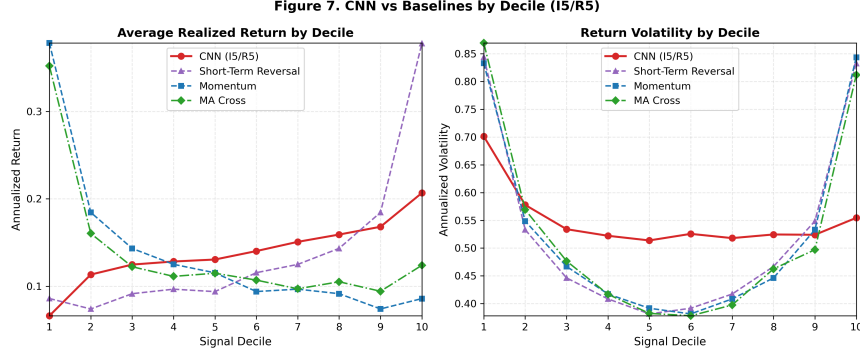


Figure 4: CNN deciles against momentum, moving-average cross, and short-term reversal baselines at the $I = 5$ horizon. Reproduction of Figure 7 of the original paper. The reversal baseline's D10 reaches an annualised return of 37.78%, against which the CNN's monotone ranking is more useful as a cross-sectional signal than as a stand-alone strategy. Analogous plots for $I = 20$ and $I = 60$ are in Appendix E.

Figure 4 shows that at $I = 5$ the CNN ranking strictly dominates both momentum and moving-average cross across the cross-section, but is itself dominated in absolute level by the short-term reversal baseline whose D10 annualised return reaches 37.78% (driven by the microstructure-style bid–ask bounce on which it relies). At $I = 20$ the momentum and moving-average curves display the same U-shaped pattern noted above, and the CNN's monotone ranking is qualitatively closer to the monthly-frequency pattern the original paper emphasises when comparing CNN deciles to traditional signals. At $I = 60$ the CNN ranking is nearly horizontal, consistent with the well-documented attenuation of price-trend signals at quarterly horizons (see Appendix E).

5.3 Cumulative returns

Figure 5 plots the cumulative log return of the H–L spread. The $I = 5$ net curve climbs steadily because the high-frequency ranking is sharp enough to absorb the transaction-cost drag. At $I = 20$ and $I = 60$, the gross curves are mildly positive but the net curves are substantially attenuated: with turnover at 84.08% and 86.68% per period (Table 2), the 20 bp round-trip cost erases most of the gross spread.

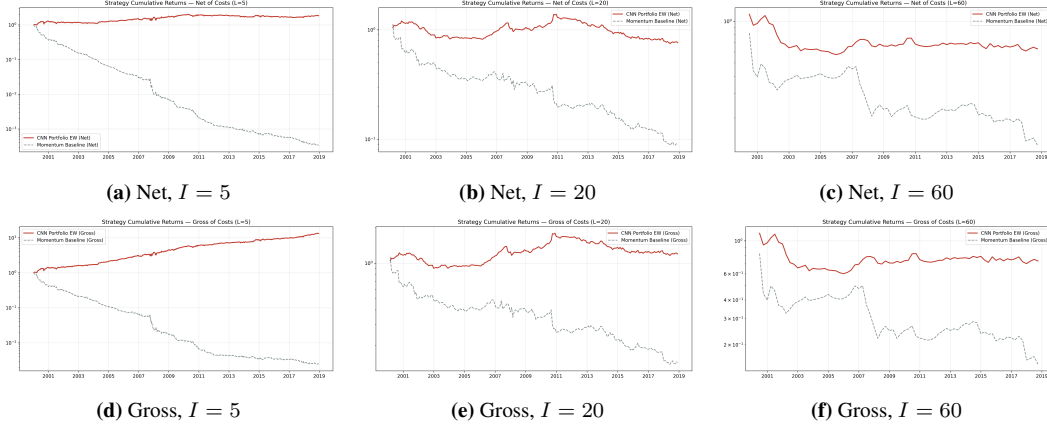


Figure 5: Cumulative log returns of the H–L decile spread, net of transaction costs (top row) and gross (bottom row). The $I = 5$ spread is the only one whose net curve still rises persistently; the $I = 20$ and $I = 60$ net curves flatten relative to their gross counterparts because round-trip turnover is in the 84–87% range (Table 2).

Table 2: Equal-weighted decile portfolio statistics (reproduction of Table 1 in the original paper). *Ret* is annualised mean return, *Std* annualised standard deviation, *SR* the resulting Sharpe ratio. “H–L” columns are for the long–short decile spread net of 20 bp round-trip transaction costs.

	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	H–L
$I = 5$ (weekly rebalance, turnover 0.858)											
Ret	0.07	0.12	0.13	0.13	0.14	0.15	0.16	0.16	0.17	0.21	0.14
Std	0.13	0.16	0.17	0.17	0.18	0.18	0.19	0.19	0.20	0.21	0.11
SR	0.55	0.75	0.77	0.77	0.75	0.79	0.83	0.85	0.89	1.04	1.29
$I = 20$ (monthly rebalance, turnover 0.841)											
Ret	0.10	0.12	0.13	0.12	0.11	0.12	0.11	0.11	0.11	0.11	0.01
Std	0.17	0.18	0.18	0.18	0.18	0.18	0.17	0.17	0.17	0.17	0.08
SR	0.59	0.63	0.72	0.65	0.63	0.65	0.65	0.66	0.68	0.68	0.16
$I = 60$ (quarterly rebalance, turnover 0.867)											
Ret	0.13	0.13	0.13	0.14	0.13	0.12	0.13	0.13	0.12	0.12	−0.01
Std	0.22	0.20	0.20	0.19	0.20	0.19	0.18	0.18	0.17	0.16	0.11
SR	0.60	0.65	0.67	0.71	0.65	0.67	0.69	0.73	0.73	0.77	−0.11

5.4 Decile portfolio statistics

Table 2 confirms the pattern visible in Figures 3–5: only the $I = 5$ horizon delivers a robust net H–L spread (annualised SR 1.29). At $I = 20$ the H–L SR is 0.16, and at $I = 60$ the spread is mildly negative (−0.11). The monotone increase in within-decile Sharpe ratios at $I = 60$ (from 0.60 at D1 to 0.77 at D10) shows that the CNN *ordering* is still informative, but the gross spread is too small relative to volatility to survive transaction costs.

5.5 Classification metrics

Table 3 shows that AUC rises with horizon, in line with the original paper, which argues that longer look-back windows expose more learnable structure; accuracy at $I = 5$ is *below* the 50% benchmark, which is consistent with a left-skewed label distribution (more days lose money in the weekly rebalance universe) rather than a strictly negative signal: the H–L Sharpe of 1.29 shows that the predicted-probability *ranking* is far more informative than the binary forecast.

5.6 Fama–French five-factor + momentum spanning

The regression in Table 4 delivers three messages. First, at $I = 5$ the strategy’s annualised alpha is statistically indistinguishable from zero (+0.28%, $t = 0.13$): the high raw Sharpe is largely captured by factor exposures (significant loadings on Mkt–RF, SMB, HML, and an opposite-signed load on

Table 3: Out-of-sample binary classification metrics for the CNN ensemble (2001–2019). N_{test} is the number of rebalance-day observations scored by the five-seed ensemble.

Horizon	Accuracy	AUC–ROC	N_{test}
$I = 5$	49.45%	0.5103	6,378,408
$I = 20$	52.68%	0.5242	1,441,167
$I = 60$	53.70%	0.5288	453,847

Table 4: Fama–French five-factor plus momentum spanning regressions for the net CNN long–short spread. The dependent variable is $r_t^{\text{CNN,LS}} - r_t^f$ regressed on $\{\text{Mkt-RF, SMB, HML, RMW, CMA, Mom}\}$ at the matched rebalance frequency. Alphas are annualised; t -statistics are in parentheses.

	$I = 5$	$I = 20$	$I = 60$
α (annualised, %)	+0.28 (0.13)	−4.30 (−2.23)	−6.65 (−3.77)
$\beta_{\text{Mkt-RF}}$	0.29 (14.24)	0.10 (2.21)	0.08 (1.24)
β_{SMB}	0.14 (3.81)	0.01 (0.11)	−0.14 (−1.24)
β_{HML}	0.13 (3.57)	0.02 (0.27)	0.23 (2.21)
β_{RMW}	0.02 (0.41)	0.32 (3.48)	0.69 (5.24)
β_{CMA}	−0.25 (−4.20)	−0.11 (−0.98)	−0.29 (−2.01)
β_{Mom}	−0.03 (−1.51)	−0.00 (−0.05)	0.23 (3.86)
R^2	0.34	0.07	0.63
N periods	989	226	74

CMA). Second, at $I = 20$ and $I = 60$ the alphas are *negative* and significant (−4.30%, $t = -2.23$; −6.65%, $t = -3.77$); the loadings on RMW are positive and large, and at $I = 60$ the strategy also picks up a positive momentum loading. Third, the very high R^2 at $I = 60$ (0.63) suggests that the CNN signal at this horizon is essentially a re-weighting of standard style factors rather than an independent source of premia.

6 Interpretability: input-gradient saliency

Following Simonyan et al. [2014], we compute the absolute pixel-level gradient of the “up” logit with respect to each input image for three high-confidence positive examples drawn from the $I = 60$ test window (Figure 6). Across all three examples the saliency mass is concentrated on the right-most 10–20% of the OHLC panel and on volume bars near sharp price moves. This is consistent with a data-driven momentum reading: the CNN essentially learns to up-weight the most-recent observations, which is also the regime in which classical price-trend signals concentrate their power.

7 Discussion and conclusion

We recover the qualitative shape of the original paper at $I = 5$ – a monotone decile ranking, a high gross-of-cost H–L spread that survives transaction costs into a 1.29 net Sharpe ratio, and a decile-by-decile dominance over momentum and moving-average cross. At $I = 20$ and $I = 60$ the qualitative ranking is preserved (Table 2, Figure 3), but the H–L spread is too thin to survive costs, and the Fama–French six-factor regression delivers a statistically significant *negative* alpha at both horizons.

Several factors plausibly account for numerical differences relative to the original paper. (i) Our CRSP-style extract, liquidity filters, and ETF-based calendar construction need not coincide exactly with the original paper’s. (ii) We aggregate five randomly seeded models; the original paper’s headline results pool a larger ensemble, which can stabilise probability estimates. (iii) Small implementation choices—chart rasterisation, float rounding, and how turnover maps into proportional costs—can move Sharpe ratios and alphas when gross spreads are thin.

The saliency analysis (Figure 6) shows that the $I = 60$ CNN learns a momentum-like reading rather than a more exotic pattern; this is consistent with a model whose alpha is largely spanned by RMW and Mom. Promising directions for future work include regime-conditioned heads, industry- or size-

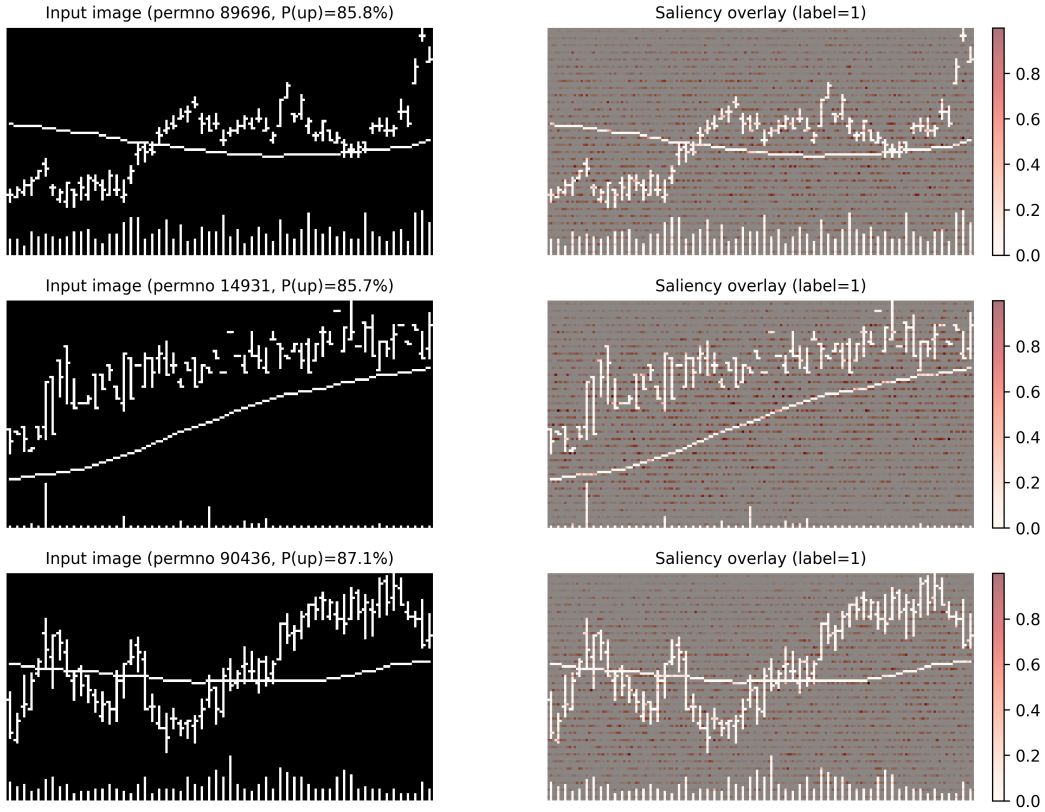


Figure 6: Input-gradient saliency for three high-confidence positive predictions of the $I = 60$ ensemble. Left column: the input chart (inverted to a white background for readability). Right column: the same image with the gradient magnitude $|\partial f_{\text{up}}/\partial x|$ overlaid in red. The network’s attention concentrates on the right-most third of each window (i.e. the most-recent price action) and on volume bars near regime changes.

neutralised label construction, and a mixture-of-experts ensemble that explicitly allocates capacity across horizons.

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A Reproduction artefact map

Every numerical claim in this paper traces back to a file in `final_submit_version/outputs/`:

- Figure 1: produced by `scripts/pipeline_split/export_paper_assets.py`.
- Figure 3, Figure 4: PNGs in `pipeline_runs/figures/`; decile-level CSVs in `pipeline_runs/tables/fig{6,7_simple}_I{5,20,60}_{cnn,ma,mom,reversal}.csv`.
- Figure 5: PNGs in `ppt_images/ppt15_cumulative_returns{,_gross}_I{5,20,60}.png`.
- Table 2: `pipeline_runs/tables/table1_I{5,20,60}.csv`.
- Table 3: `ppt_images/ppt16_classification_metrics.csv` plus shape of `cache/preds_I{5,20,60}.npy`.
- Table 4: produced by `scripts/pipeline_split/export_ff_regression.py`; intermediate file `pipeline_runs/tables/ff_regression_combined.csv`.
- Figure 6: produced by `scripts/pipeline_split/export_saliency_map.py` using the $I = 60$ checkpoint at seed 42.

B Hyperparameters and per-block tensor shapes

Table 5 reports the per-block configuration used in the three CNN backbones; all blocks share the `Conv→BatchNorm→LeakyReLU(0.01)→MaxPool(2, 1)` pattern.

Table 5: CNN backbone configuration by horizon. “–” denotes a convolutional block not used for that horizon. FC input dimensions match Figure 2.

Block	$I = 5$	$I = 20$	$I = 60$
Block 1 channels	$1 \rightarrow 64$	$1 \rightarrow 64$	$1 \rightarrow 64$
Block 1 stride	(1,1)	(3,1)	(3,1)
Block 1 dilation	(1,1)	(2,1)	(3,1)
Block 2 channels	$64 \rightarrow 128$	$64 \rightarrow 128$	$64 \rightarrow 128$
Block 3 channels	–	$128 \rightarrow 256$	$128 \rightarrow 256$
Block 4 channels	–	–	$256 \rightarrow 512$
FC input dim	15,360	46,080	184,320
Approx. params	~ 150 k	~ 700 k	~ 3.0 M

C Computational footprint

The cleaned daily panel occupies approximately 11.1 GB. Per-seed model checkpoints are approximately 0.6 MB ($I = 5$), 2.8 MB ($I = 20$), and 11.8 MB ($I = 60$); five seeds per horizon give a total of about 76 MB. Training and ensemble inference were performed on a single CUDA GPU (with an Apple-silicon fallback for development); end-to-end execution of the pipeline takes on the order of a few wall-clock hours once the cleaned panel has been constructed.

D Decile return plots for $I = 20$ and $I = 60$

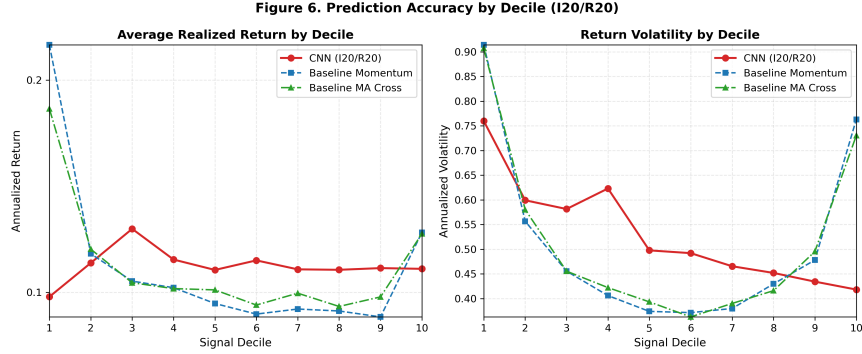


Figure 7: Annualised decile returns at the $I = 20$ horizon for the CNN, matched-horizon momentum, and moving-average cross signals. Reproduction of Figure 6(b) of the original paper.

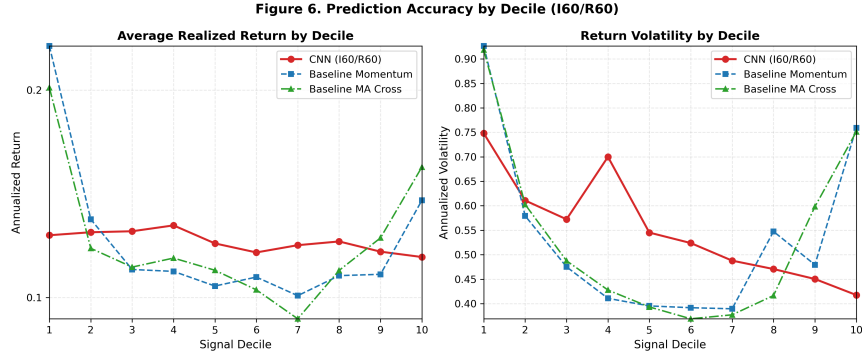


Figure 8: Annualised decile returns at the $I = 60$ horizon for the CNN, matched-horizon momentum, and moving-average cross signals. Reproduction of Figure 6(c) of the original paper. The CNN ordering is nearly flat across deciles at this horizon, consistent with the attenuation of price-trend signals at quarterly look-back windows.

E Baseline comparison plots for $I = 20$ and $I = 60$

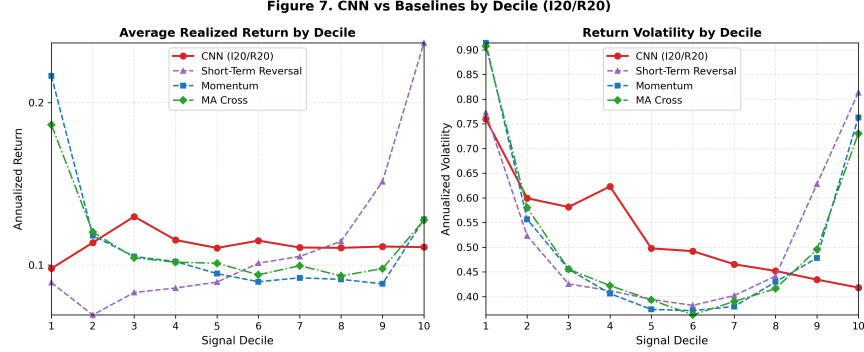


Figure 9: CNN deciles against momentum, moving-average cross, and short-term reversal baselines at the $I = 20$ horizon. Reproduction of Figure 7 of the original paper. The CNN’s monotone ranking is qualitatively closer to the monthly-frequency pattern than at $I = 5$.

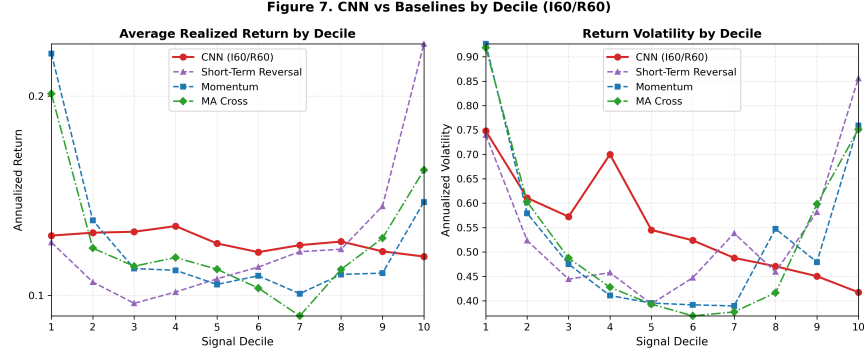


Figure 10: CNN deciles against momentum, moving-average cross, and short-term reversal baselines at the $I = 60$ horizon. The CNN decile ordering is nearly flat, consistent with the attenuation of price-trend signals at quarterly look-back windows.

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